EXPERIMENT 4: RULE-BASED EXPERT SYSTEM FOR NETWORK TROUBLESHOOTING

Aim:

To infer a possible network troubleshooting action from user-provided symptoms.

Software Required: Python 3.x

Procedure / Algorithm:

1. Ask whether internet access is available.
2. Ask whether Wi-Fi is connected.
3. Apply IF-THEN rules to display a possible troubleshooting step.

Program:

```
internet = input(
   "Can you access the internet? (yes/no): "
).strip().lower()


wifi = input(
   "Is Wi-Fi connected? (yes/no): "
).strip().lower()


if internet == "yes":
   print("Diagnosis: Internet connection is working.")
elif wifi == "no":
   print("Diagnosis: Check Wi-Fi settings or router power.")
else:
   print(
      "Diagnosis: Check router connectivity "
      "and network settings."
   )
```

Sample Input:

Can you access the internet? (yes/no): no

Is Wi-Fi connected? (yes/no): yes

Expected Output:

```
>>
= RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-4.py
Can you access the internet? (yes/no): YES
Is Wi-Fi connected? (yes/no): YES
Diagnosis: Internet connection is working.
>>
```

Result:

Thus, the rule-based expert system selected a troubleshooting suggestion using IF-THEN conditions.